

Leakage-free temporal protocol

H History OD

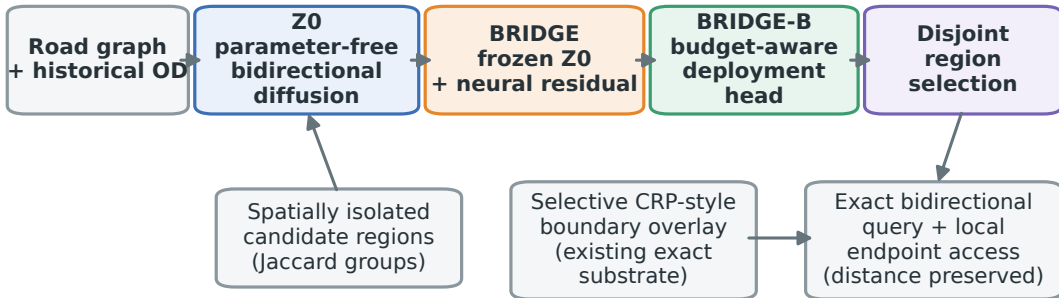
0-35% · model input

Y Current labels

35-70% · exact regional gain

F Future test

70-100% · frozen evaluation



Learning allocates offline materialization resources; online shortest-path answers remain exact.